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CENTER FOR QUANTUM & TOPOLOGICAL SYSTEMS

Theory Guided Features and Architecture Search for Predicting Hyperbolic Volume from Jones Polynomials

A phase evaluation feature that breaks the J_2 ceiling with the adjoint J_3, using networks under 3,000 parameters

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Motivation and Objectives

What is a Jones polynomial?

- A knot invariant: a Laurent polynomial in a variable q, built from a simple skein relation, that stays the same however you bend or stretch the knot.
- Introduced by Vaughan Jones in 1984 (a Fields Medal discovery), it opened the first real bridge between knot theory and physics.
- It links knots to quantum field theory and statistical mechanics; its colored versions J_2, J_3, … come from higher dimensional representations and carry progressively more information.
- Powerful but not complete: it usually tells two knots apart, yet some distinct knots still share the same polynomial.

Task

- Predict a knot's hyperbolic volume from its Jones polynomial.
- Volume is an exact geometric invariant: every hyperbolic knot complement carries a unique hyperbolic metric (Mostow rigidity), computed here with SnapPy.
- The Jones polynomial is combinatorial and quantum; volume is geometric. A good fit is empirical evidence for the volume conjecture that ties the two together.

Why learn it

- The volume conjecture only bites in the infinite color limit, where evaluating J□ grows exponentially expensive and is effectively intractable.
- Low color polynomials (J_2, J_3) are cheap to compute but individually sit far from that limit.
- So we train on cheap low color data paired with exact volumes; the network learns the hidden map from low color coefficients to volume and applies it in a single forward pass.

Objectives

1. Measure the accuracy gain from adding the volume conjecture phase feature $V(q_0)$, under a matched search budget.
2. Prove by ablation (zeros vs. noise) that the gain is real signal, not just extra dimensions.
3. Compare J_2 vs. J_3 (adjoint) vs. J_2+J_3 on one knot set: does the richer invariant break the J_2 ceiling?

The volume conjecture
Kashaev 1997; Murakami–Murakami 2001

$$\lim_{r\to\infty} \frac{2\pi}{r} \log \left| J_r(K;\, e^{2\pi i/r}) \right| = \mathrm{Vol}(S^3 \setminus K)$$

Methodology

Data

- A: 12,955 hyperbolic knots up to 13 crossings, with J_2 (34 coefficients) and volume.
- B: 1,419 knots up to 12 crossings with J_2 and J_3; volumes computed in SnapPy.
- B needed cleaning (mixed conventions, framing shifts). All features use a shift invariant canonical form, validated against A.

Features

- Summary stats (span, nonzero count, L1/L2 norms, V(1), |V(−1)|), raw coefficients, and the phase evaluation [Re, Im, |·|] of $V(q_0)$.
- The angle is swept, not assumed: correlation with log volume peaks at $\theta = 13\pi/16$ for J_2 (r = 0.943) and $\theta = 5\pi/9$ for J_3 (r = 0.959).
- A single scalar already tracks volume almost as well as a full network.

Search

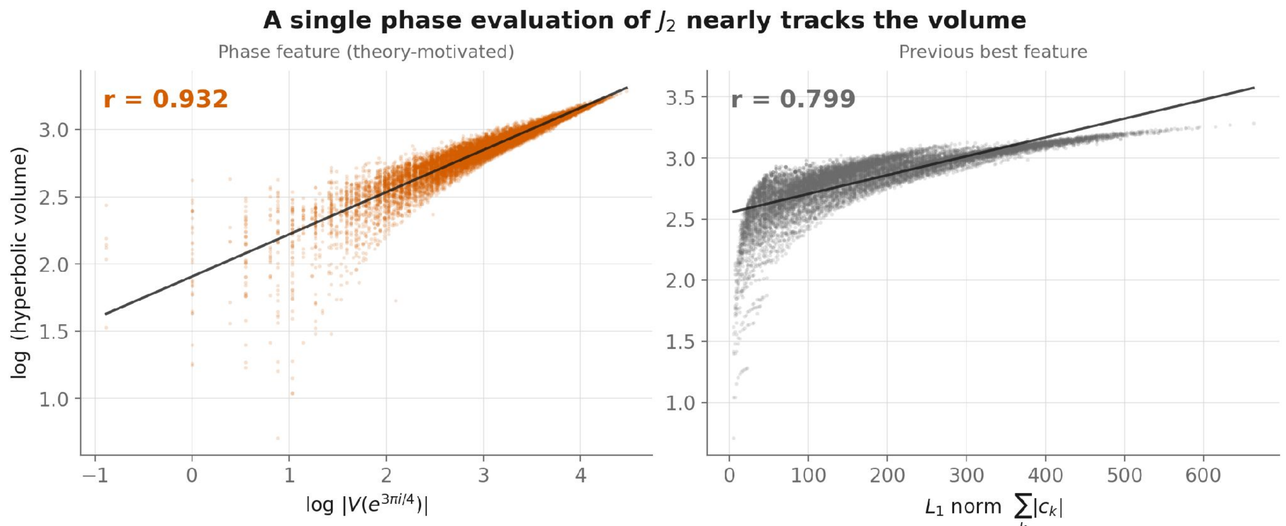
- NSGA-II multiobjective search over MLP genomes (depth, width, activation, dropout, L2, learning rate, batch).
- Objectives: validation MRE and $\log_{10}$(params); overfitting gap as a constraint.
- The Pareto knee is retrained from scratch (7 seeds, 500 epochs), then tested once on an untouched 70/15/15 split.
- Same budget every run: pop 40, 40 generations, 4 seeds. MRE = mean(|pred − true| / true); accuracy = 1 − MRE.

Controls

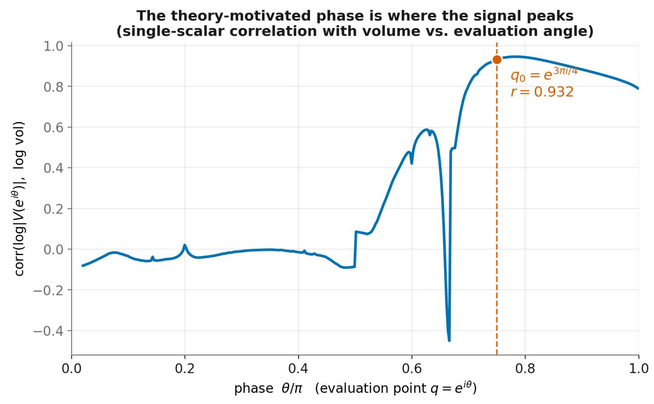
- Fix the knee and training, then replace the phase columns with zeros or Gaussian noise.
- Any change in accuracy is then purely those columns' information.

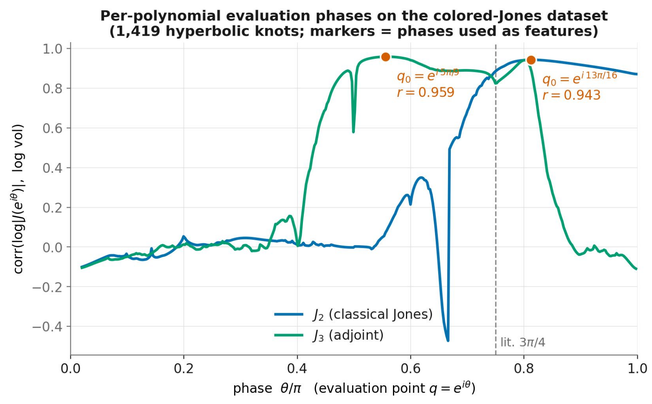
Optimal evaluation phases: ours vs. literature vs. classical root

Invariant	Ours (swept)	Literature	Classical root $e^{2\pi i/r}$
J_2 (Jones)	$e^{13\pi i/16}$ (r = 0.943)	$e^{3\pi i/4} = e^{12\pi i/16}$ (Craven et al. '21)	$e^{\pi i}$
J_3 (adjoint)	$e^{5\pi i/9}$ (r = 0.959)	$e^{8\pi i/15}$ (Hughes et al. '25)	$e^{2\pi i/3}$



Feature to volume correlations: $|V(q_0)|$ is the strongest single feature; phase sweeps peak at $\theta = 13\pi/16$ (J_2) and $\theta = 5\pi/9$ (J_3)



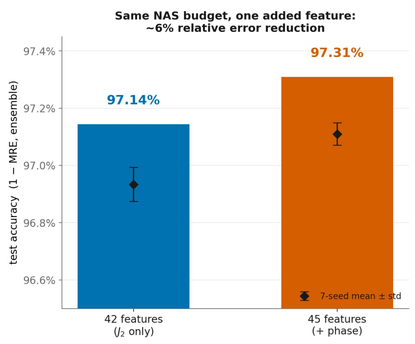


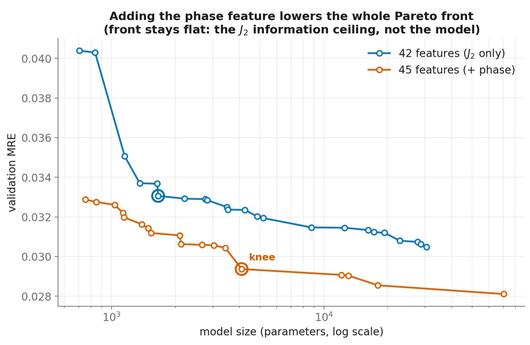
Results

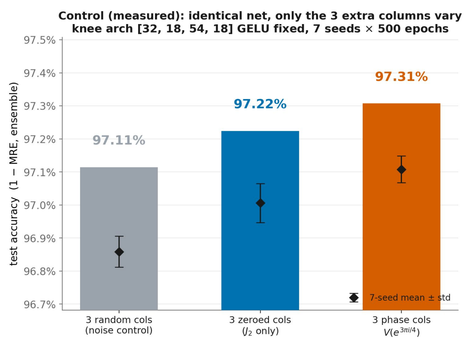
Phase feature, dataset A

97.14% → 97.31%

- Ensemble accuracy at a matched budget (test MRE 0.0307 → 0.0289, about 6% less error, ~3σ).
- The whole Pareto front shifts down.
- Ablation: phase 97.31% > zeroed 97.22% > noise 97.11%. The gain is the volume signal.



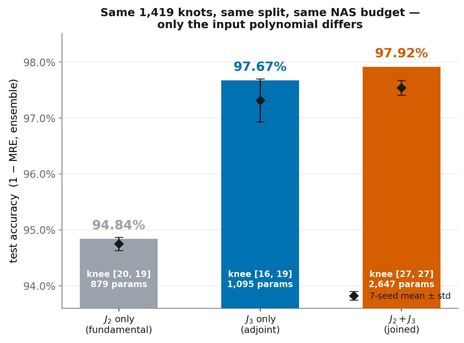


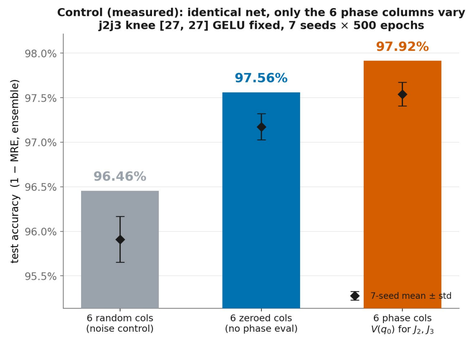


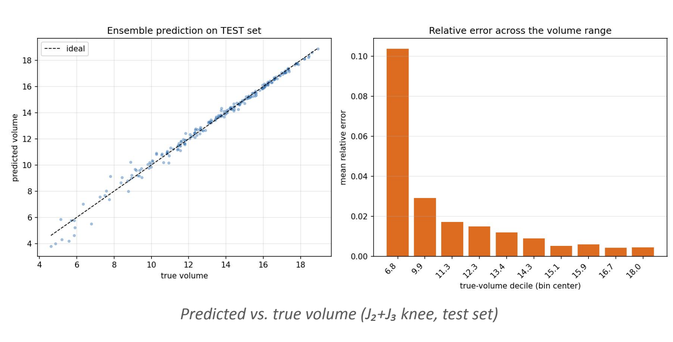
J_3 breaks the J_2 ceiling, dataset B

94.8% → 97.7% → 97.9%

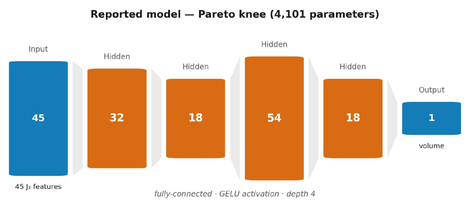
- $J_2 \to J_3 \to J_2$+J_3 ensemble accuracy, same 1,419 knots and split (test MRE 5.16% → 2.33% → 2.08%).
- J_3 alone cuts J_2's error 2.2×.
- Ablation at the fixed [27, 27] knee: phase 97.92% > zeroed 97.56% > noise 96.46%.



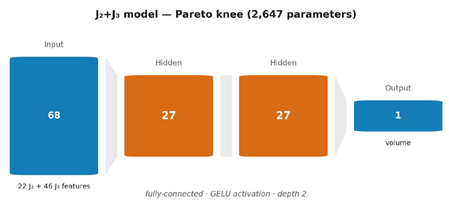




Predicted vs. true volume (J_2+J_3 knee, test set)



A knee: 45→32→18→54→18→1 GELU, 4,101 params



B knee: 68→27→27→1 GELU, 2,647 params. MRE 0.0246 ± 0.0013, ens. 0.0208

Accuracy per parameter vs. literature

- Hughes et al. 2025 (about 180k param MLPs on 177k knots: 1.85% / 0.62% / 0.40% MRE) shows the same $J_2 \to J_3 \to J_2$+J_3 ordering.
- We reproduce it with 70 to 200× smaller models on about 125× less data.
- Earlier: Jejjala, Kar, Parrikar 2019 (about 12k params, 2.45%); Craven et al. 2022 (about 22k, ~3%).
- Flat Pareto fronts and sub 3k param knees: accuracy is capped by the invariant, not model size.

Future work

- Scale J_3 to the ≤13 crossing census.
- Push higher: predict the volume signal of J_4, J_5, … from cheap low color data.
- Interpret the knee networks symbolically.

$$q_r^* = \exp\left(2\pi i \frac{r+1}{r(r+2)}\right) \longrightarrow e^{2\pi i/r}$$

matches the r = 2, 3 phases; → classical root as $1/r - 1/r^2$ — the target for J_4, J_5, …

Take-home: a single theory-guided phase evaluation carries most of the volume signal — J_3 breaks the J_2 ceiling with < 3,000-parameter networks; accuracy is capped by the invariant, not the model.

This work was supported in parts by the NYUAD Center for Quantum and Topological Systems (CQTS), funded by Tamkeen under the NYUAD Research Institute grant CG008, and the NYUAD Center for Cyber Security (CCS), funded by Tamkeen under the NYUAD Research Institute Award G1104.